UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

Lesson 7 — Trees and Ensembles

Technologies for Artificial Intelligence

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Agenda

- A fourth family, built on a threshold
- Decision trees, and why one alone is mediocre
- Depth as the bias-variance dial, again
- Bagging and random forests
- Feature importance, and its limits
- Gradient boosting
- Choosing among trees, forests and boosting

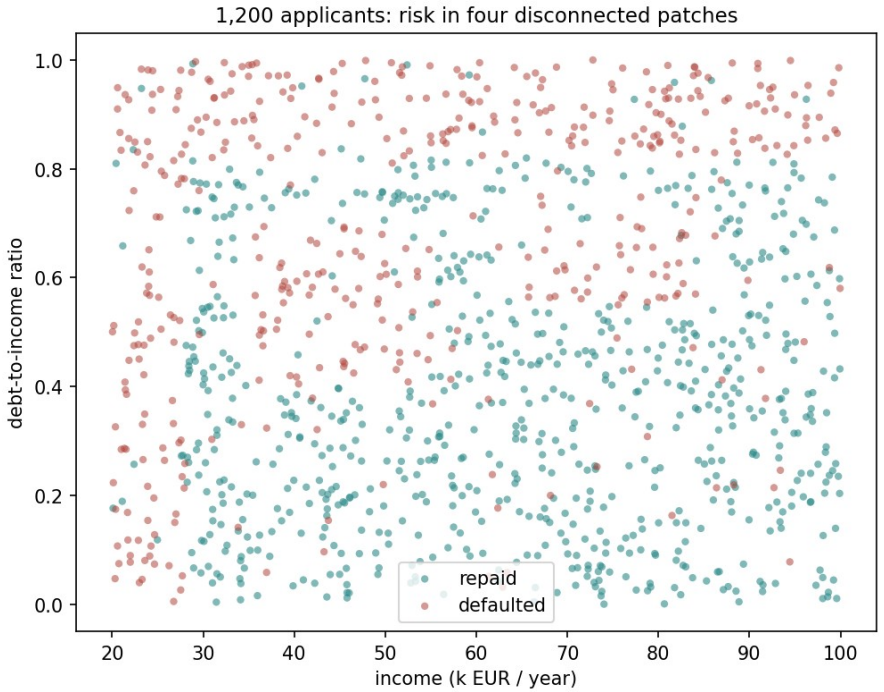
Exercise 6 returned

- Marks were for the **defence of which family**, not for the winning score
- The recurring gap: a strong accuracy number with no check of *why* the assumption behind it held
- Today’s models come with the same obligation

1,200 loan applicants, two figures each

- Income in thousands of euros a year, debt-to-income ratio
- Underwriting is a **checklist**, not a formula: an income floor, a debt ceiling, and two “stressed” combinations in between
- **7%** of labels are deliberately flipped; **38.7%** of applicants defaulted, so the majority baseline is **0.613**

Four disconnected risky regions



No straight line separates them

Model	Cross-validated accuracy
Always predict the majority class	0.613
Logistic regression	**0.748 ± 0.030**

The whole algorithm, in one sentence

- Find the single (feature, threshold) split that makes the two resulting groups **as pure as possible**
- Apply it. Repeat on each group, independently
- Stop when a stopping rule is met
- Nothing is estimated — no coefficient, no gradient with respect to a parameter vector
- **Greedy**: whichever split helps most *right now*, with no lookahead

How pure is a group of examples?

- A leaf that is all one class needs no more questions
- A leaf split evenly between classes is the worst case — a coin flip
- **Impurity** is a number that captures this, so “as pure as possible” has a target to maximise against

Gini impurity

$$G = 1 - \sum_{c=1}^{C} p_c^2$$

For two classes, one number

- G = 2p(1-p), where p is the fraction of one class
- **0** when the group is pure (p = 0 or p = 1)
- **0.5**, the maximum, at p = 0.5
- On the full 1,200 applicants, p = 0.387 gives G = 0.474

The gain a split buys

- A candidate split divides a group into a left and a right child
- It is worth taking if the children are, **on average, purer** than the parent was
- “On average” is weighted by how many examples land in each child — a split that purifies nine points and dumps one troublemaker into its own leaf is not free

The reduction in impurity

$$\Delta G = G(\text{parent}) - \left(\frac{n_L}{n}\, G(\text{left}) + \frac{n_R}{n}\, G(\text{right})\right)$$

Scanning for the best split

- For a continuous feature, scan **every midpoint** between adjacent sorted values as a candidate threshold
- Do this for **every feature**; keep the best pair
- Repeat the search inside each child
- **Greedy**: Δ G is judged one split at a time, never jointly

Worked example: the root split

- p = 0.387 defaulted, so the parent group’s impurity is **0.474**
- Exhaustive search finds `debt_ratio <= 0.82` as the best root split
- Recognisably the **debt ceiling** the data was built with

Checked against scikit-learn

- The from-scratch implementation agrees with scikit-learn's `DecisionTreeClassifier` to the **fourth decimal place** of resulting accuracy, at every depth tested
- Not a coincidence: both search the same space by the same greedy rule
- From here on, the notebooks use scikit-learn's implementation — the same algorithm, compiled, not a different one

What greedy misses

- The search never asks “which pair of splits helps most together” — only “which single split helps most right now”
- A feature that only pays off **in combination** with a later split can be invisible to a shallow tree
- Exactly what happens to this dataset’s two interior “stressed” islands — covered next

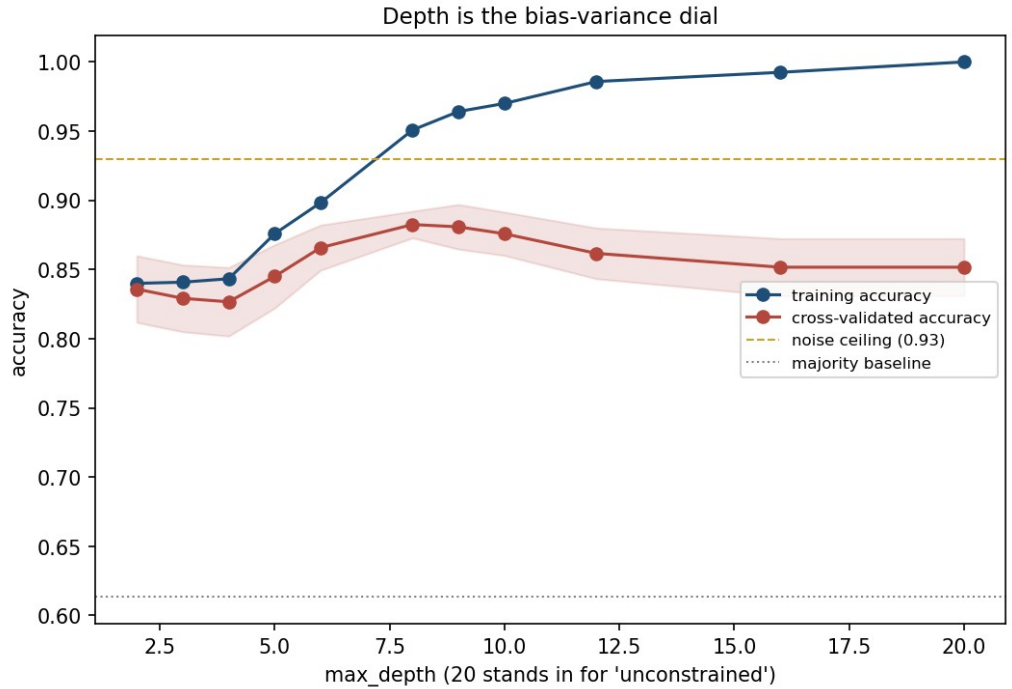
Depth is the bias-variance dial

- `max_depth`: how many questions a tree may ask before a leaf must stop and vote by majority
- **Shallow tree** — few questions, only the coarsest structure. High bias, low variance
- **Deep tree** — enough questions to isolate almost any subset, down to single points. At full depth, training accuracy is **exactly 1.000 on any dataset**, including one with no structure at all

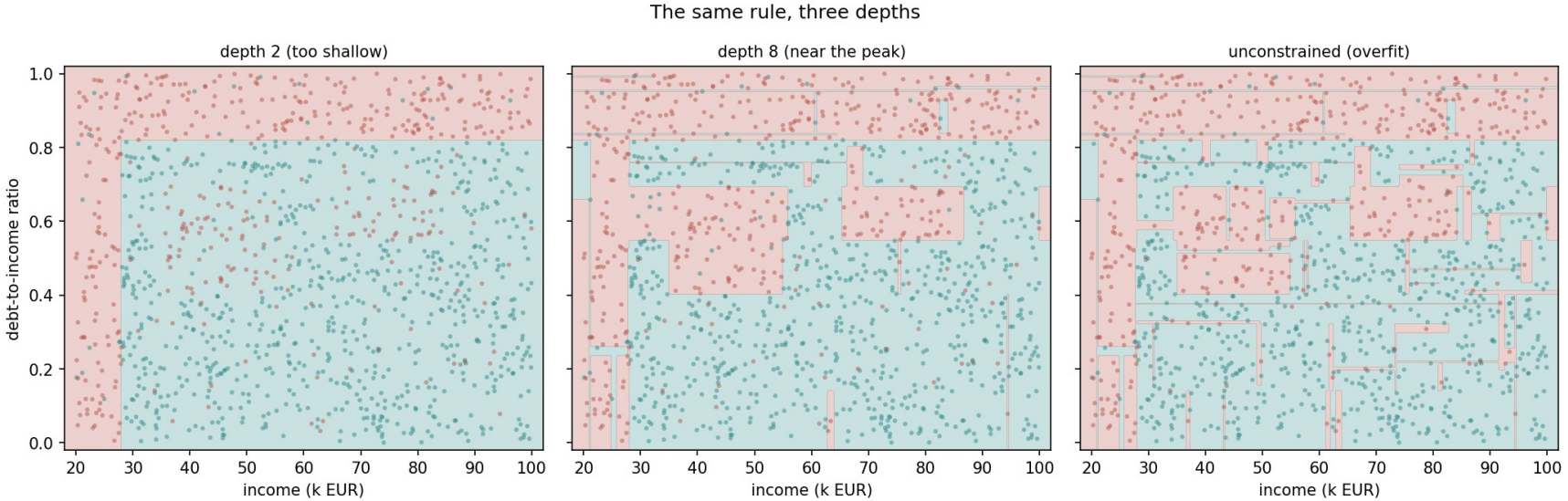
Measured, at seven depths

max_depth	Training accuracy	Cross-validated accuracy
2	0.840	0.836 ± 0.024
4	0.843	0.827 ± 0.025
6	0.898	0.866 ± 0.016
8	0.951	**0.882 ± 0.010**
12	0.986	0.862 ± 0.018
16	0.993	0.852 ± 0.021
unconstrained	**1.000**	0.852 ± 0.021

The curve bends downward



The same rule, three depths



Reading a tree, and the knob that keeps it small

- `max_depth` grows the tree **breadth-first** — every leaf at level d splits before any leaf reaches d+1, useful or not
- `max_leaf_nodes` grows it **best-first** — always expanding whichever leaf offers the largest Δ G
- With `max_leaf_nodes = 9`: **90.8%** training accuracy, every threshold recognisable — 28 the income floor, 0.82 the debt ceiling

A tree that doesn’t stay still

- Two trees at `max_depth = 6`, fit to independent 65% resamples of the same 1,200 applicants
- The debt ceiling — the strongest split — agrees to within **0.001**
- The trees grow **31 and 33 leaves**, and **disagree on 11.7%** of predictions on the full dataset
- The strongest split is stable; the many weaker ones are not, and they decide most of a tree’s shape

Notebook 1, live

- Decision trees from scratch, checked against scikit-learn
- Depth swept, the two boundaries drawn, and the instability measured

Break

- Twelve minutes

Bagging: averaging away the instability

- If a tree's mistakes are somewhat random — different across resamples, as just measured — **many trees, averaged**, should partly cancel them out
- Draw a **bootstrap sample**: m rows chosen **with replacement** from the m training rows
- Fit a tree to it. Repeat `n_estimators` times; predict by **majority vote**
- The tree algorithm itself is unchanged — only what surrounds it is new

How much data one bootstrap sample sees

- Each bootstrap sample is drawn from the same m rows, with replacement
- About **63.2%** of the distinct rows are drawn at least once
- The remaining **36.8%** are left out entirely — called **out-of-bag (OOB)**
- At m = 1,200: a single simulated draw left out 36.6% of rows, close to the limit

Out-of-bag: free validation

- Every out-of-bag row is validation data for the tree that missed it
- Averaging them gives the **OOB score**, at no extra split
- A 300-tree forest: OOB **0.9117**, 5-fold cross-validated **0.9117 ± 0.021**
- That match to four decimals is **this seed's luck**: over twelve seeds the gap is 0.003 — agreement to within their own noise, not to every decimal

Why averaging reduces variance

- Model each tree’s prediction as noisy, with variance σ^2, and let ρ be the correlation between any two trees
- Averaging B trees drives variance toward a **floor** of $\rho\sigma^2$, **not zero** — adding trees past that point buys almost nothing
- Averaging **cannot fix bias**: unbiased-but-noisy trees average to unbiased; trees sharing a systematic error keep that error exactly

Bagging, measured

Model	Mean test accuracy	Standard deviation
Single unconstrained tree	0.856	0.0185
Bagging, 100 trees	**0.902**	**0.0153**

Random forests: decorrelating the trees

- The variance floor is $\rho\sigma^2$ — lowering it further means making the trees agree **less**
- Bagged trees still see **every feature** at every split, so a strong feature pulls most of them towards splitting on it first
- A **random forest** restricts each split to a random subset of `max_features` (default $\sqrt{n}$, rounded down) — different trees see different features at the same point, lowering ρ

Random forests, measured

Model	Mean test accuracy	Standard deviation
Single unconstrained tree	0.856	0.0185
Bagging, 100 trees	0.902	0.0153
Random forest, 100 trees	**0.908**	**0.0137**

Feature importance: what the forest claims to tell you

- A random forest reports **feature importance**: total impurity reduction each feature is responsible for, summed and averaged across every tree and every split
- Tempting to read as “the forest tells you which features matter”
- With only the two real features in this dataset, it does exactly that
- The question: what happens once most of the columns carry **nothing**?

More columns, more noise

Noise columns added	Total columns	Cross-validated accuracy	Importance on noise
0	2	0.912	0.0%
5	7	0.874	33.9%
20	22	0.837	**54.5%**

Notebook 2, live

- Bagging and random forests, from scratch and against scikit-learn
- The out-of-bag score checked against cross-validation
- The noise-column experiment, run live

Gradient boosting: correcting mistakes in sequence

- Bagging and random forests build trees **independently** and average at the end
- **Boosting** builds trees **in sequence**: each new tree is fit specifically to correct what the trees so far got wrong
- Each addition is a shallow tree — depth 2 or 3, a **weak learner**
- The learning rate α sets **how much** of each new tree’s correction is actually applied

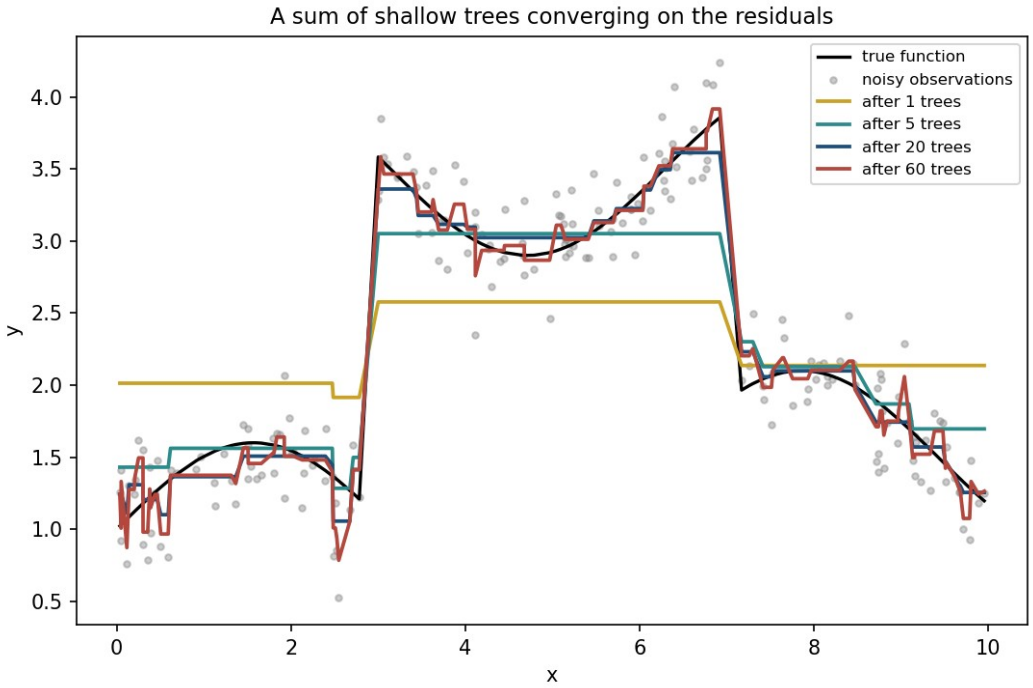
Building the ensemble, one tree at a time

$$F_0(x) = \bar{y}, \qquad F_t(x) = F_{t-1}(x) + \alpha \cdot h_t(x)$$

What each tree is fit to

- For squared-error regression, “what remains unexplained” has an exact name: the **residual**, y - $Ft_{-1}(x)$
- This is a special case of a more general idea — **functional gradient descent**: ht approximates the negative gradient of the loss with respect to Ft_{-1}
- For squared error, “fit the residual” and “fit the negative gradient” coincide exactly

A sum of shallow trees converging on the residuals



Reading the picture

- **1 tree**: mean squared error (MSE) against the true function 0.395 — barely moves the flat guess
- **5 trees**: MSE 0.068 — the coarse shape appears
- **20 trees**: MSE **0.012** — the ensemble’s best point
- **60 trees**: MSE rises back to 0.018 — tracking noise, not signal

Classification: the same idea, a different gradient

- The label y is 0 or 1; the ensemble’s output becomes a probability, $p = \sigma(F)$
- The loss is **log-loss** — lesson 4’s loss for logistic regression
- The negative gradient, the **pseudo-residual** each tree fits, is y - p
- True label minus predicted probability: lesson 4’s error term exactly

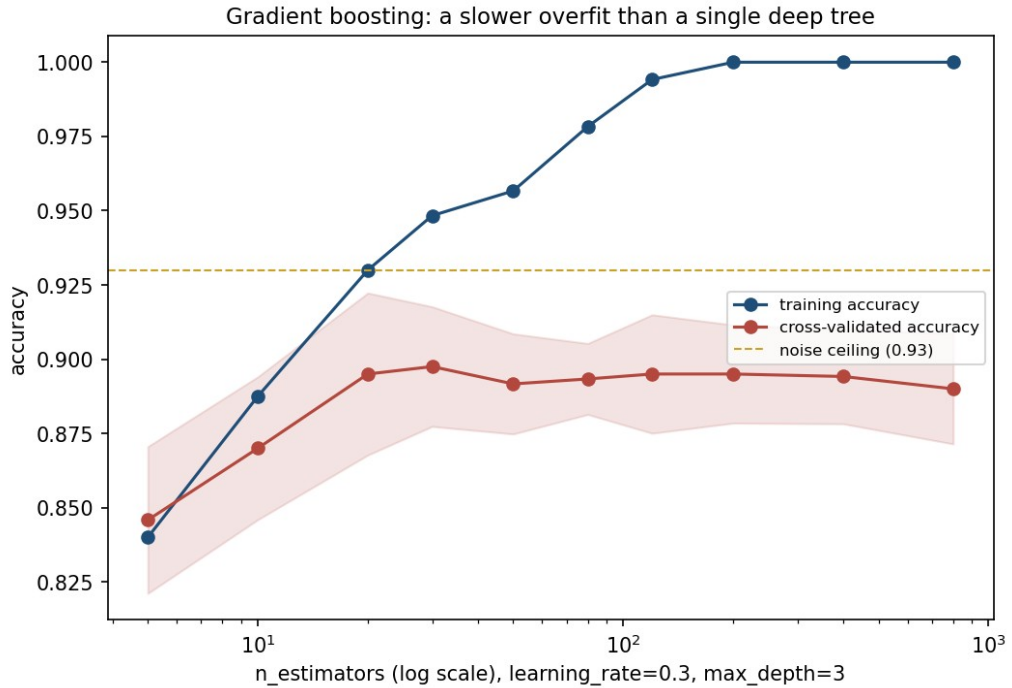
On the loan data

- `GradientBoostingClassifier`, defaults: **100 trees**, learning rate $\alpha = 0.1$
- Reaches **0.902 ± 0.020** cross-validated accuracy
- Already ahead of any single tree from earlier in the lesson

Boosting never stops fitting on its own

`n_estimators` (α = 0.3)	Training accuracy	Cross-validated accuracy
5	0.840	0.846 ± 0.025
10	0.888	0.870 ± 0.024
20	0.930	0.895 ± 0.027
30	0.948	**0.898 ± 0.020**
80	0.978	0.893 ± 0.012
200	1.000	0.895 ± 0.017
800	1.000	0.890 ± 0.019

Training climbs to 1.000 and stays; the honest curve turns over first



Fixing 120 trees, varying only the learning rate

Learning rate α	Training accuracy	Cross-validated accuracy
0.02	0.888	0.872 ± 0.025
0.05	0.914	0.897 ± 0.022
0.10	0.939	**0.902 ± 0.022**
0.30	0.994	0.895 ± 0.020
1.00	1.000	0.878 ± 0.020

Notebook 3, live

- Boosting from scratch on residuals, and against scikit-learn
- The classification pseudo-residual, and the learning-rate sweep

Every method, on the same five folds

Model	Cross-validated accuracy
Majority baseline	0.613
Logistic regression	0.748
Single tree, unconstrained	0.852
Single tree, `max_depth = 8`	0.883
Gradient boosting, 30 trees	0.898
Bagging, 100 trees	0.904
Random forest, 100 trees	**0.911**
noise ceiling	*≈ 0.93*

Three findings

- Every ensemble lands within about a point and a half of the others — **which strategy** you pick matters far less than **whether you ensemble at all**
- The single tuned tree is not far behind, at 0.883 against 0.911, and it is the only model on the table a person could read start to finish
- The **unconstrained tree is the worst model on the list**, despite being the most flexible one

How to choose, in practice

Situation	Reach for
The decision must be explained to the person it affects	A single tree, depth-tuned, or nothing on this list
Tabular data, accuracy matters most, no explanation required	A random forest, as a strong and low-effort default
Training time matters, or the signal needs sequential refinement	Gradient boosting, tuned on a validation set
Very high-dimensional, few informative columns	Any tree method, cautiously
Rows only, and OOB score suffices for validation	A random forest — no explicit train/test split needed

54% of the importance landed on noise

- 20 noise columns, 2 real features — not a small effect, and not a bug
- The same restriction that **decorrelates** the trees occasionally hands a split to noise instead
- Averaging reduces **variance of the prediction**; it does **not** guarantee small importance for irrelevant columns
- The fix: cross-check against permutation importance on held-out data

What to take away

- A tree splits greedily on Gini impurity, verified to the **fourth decimal place** against scikit-learn
- Depth is the bias-variance dial: accuracy peaked at **depth 8**, three points above the unconstrained tree
- Bootstrap resampling leaves **36.8%** of rows out-of-bag, free validation
- **54% of a random forest’s importance landed on 20 noise columns** — today’s number worth remembering

Homework

- **Exercise 7**, due **Friday 13 November 2026, 23:59**
- `Exercises/07_trees_and_ensembles.md`